Ecuaciones Principales.

$$Ve(t) = Ri_0(t) + L\frac{d[i_2(t) - i_2(t)}{dt} + R[i_1(t) - i_2(t)]$$

$$V_s(t) = Ri_2(t) + \frac{1}{C}\int i_2(t)dt$$

- Transformada de Laplace

$$Ve(s) = RI_1(s) + Ls[I_1(s) - I_2] + R[I_0(s) - I_2(s)$$

$$V_s(s) = RI_2 + \frac{I_2(s)}{Cs}$$

- Proc. Algebraico

$$Ve(s) = (Ls + 2R)I_1 - (Ls + R)I_2(s)$$

$$V_s(s) = \left(R + \frac{1}{Cs}\right)I_2(s) = \frac{CRs+1}{Cs}I_2(s)$$

$$\frac{3(CLRs^2 + Cs(CR+1)s + 2R}{CLs^2 + CRs}$$

- Func. transferencia:

$$\frac{V_s(s)}{Ve(s)} = \frac{\frac{CR+1}{Cs}}{\frac{3CLRs^2 + [sCR+L]s + 2R}{CLs^2 + CRs}}$$

$$\frac{V_s(s)}{Ve(s)} = \frac{CLRs^2 + (CR^2 + L)s + R}{3CLRs^2 + Cse R + 2 + 2R.}$$

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Error de estado est.

$$e(s) = \lim_{s \to s_0} \left[1 - \frac{CLB\cos^2 + C(LB+L)\cos + 1B)}{3(CLB)\cos^2 + (5CB+L)\cos + 2B} \right.$$

$$e(s) = \lim_{s \to 0} \quad 1 - 0.05 \pm 0.5$$

$$e(s) = 0.5\,V$$

- Estabilidad L. Abierto.

$a = 3CLB$ $\quad B = 2.2\,K$
$b = 5CB^2 + L$ $\quad C = 470\,\mu F$
$c = 2B$ $\quad L = 68\,mH$

$$s_{1,2} = \frac{-(5CB^2+L) \pm \sqrt{(5CBAL)^2 - 4(3CLB)(2B)}}{2(3(LB))}$$

$s_1 = 0.3868476579$

$s_2 = -53921.50415.$ Sobreamortiguado

- Modelo de EC. integrodiferenciales

$$V_e(t) = 2i_1(t) + L\,d\left[\frac{i_1(t) - i_2(t)}{dt}\right] - Bi_2(t)$$

$$i_2(t)\left[\frac{i_1(t) - i_2(t)}{dt}\right] + Ri_2(t) = Ri_2(t) + R_1 2(t) + Vi_2(t) + \frac{1}{C}\int i_2(t)\,dt$$

$$I_1(t) = \int L\,dE$$

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F. transferencia "Ec. Principales"

$$F_a(t) = f_z(t) + F_L(t) = f_c(t) + f_R(t)$$

$$f_z(t) = \frac{P_a(t) - P_p(t)}{Z}$$

$$F_L(t) = \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt$$

$$F_c(t) = C\frac{dP_p(t)}{dt}$$

$$P_p(t) = \frac{P_p(t)}{R}$$

"Transformada de Laplace"

$$F_a(s) = f_z(s) + F_L(s) = F_c(s) + F_R(s)$$

$$f_z(s) = \frac{P_a(s) - P_p(s)}{Z}$$

$$F_L(s) = \frac{P_a(s) - P_{ps}(s)}{Ls}$$

$$F_c(s) = Cs\,P_p(s)$$

$$f_R(s) = \frac{P_p(s)}{R}$$

"Procedimiento algebraico"

$$f_z(s) + F_L(s) = F_c(s) + f_R(s)$$

$$\frac{P_a(s) - P_p(s)}{Z} + \frac{P_a(s) - P_p(s)}{Ls} = Cs\,P_p(s) + \frac{P_p(s)}{R}$$

$$\frac{P_a(s)}{Z} - \frac{P_p(s)}{}$$

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"Resultado:"

$$\frac{P_2(s)}{P_1(s)} = \frac{LBs + HBz}{CLBz s^2 + L(z+HB)s + Bz}$$

Estabilidad del sistema a lazo abierto:

$$CLBz s^2 + L(z+HB)s + Bz = 0$$

$$a = CLBz$$
$$b = L(z+HB)$$
$$C = Bz$$
$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2-4ac}}{2a} = \frac{-L(z+HB) \pm \sqrt{L^2(z+HB)^2 - 4(CLB)^2 z^2}}{2CLBz}$$

"Hipotesis"

$z = 0.020$ 0.01
$C = 0.250$
$B = 0.600$
$L = 0.005$

$\lambda_1 = -3.9463$
$\lambda_2 = -202.72$

* Sistema presenta respuesta estable a un escalón unitario de entrada

$z = 0.01$
$L = 0.001$
$B = 1.000$
$C = 3.500$

$\lambda_1 = -14.4286 + 8.8052i$
$\lambda_2 = -14.4286 - 8.8052i$

y ~~sobreamortiguada~~ Subamortiguada

"Normotenso"

$z = 0.033$
$C = 1.500$
$B = 0.950$
$L = 0.010$

$\lambda_1 = -3.9289$
$\lambda_2 = -16.977$

$z = 0.01$
$L = 0.001$
$B = 1.00$
$C = 1.500$

$\lambda_1 = -12.0616$
$\lambda_2 = -55.2717$

* ~~Mismo caso que hipotenso~~

El sistema presenta una respuesta estable y sobreamortiguada a un escalón unitario.

"Hipertenso"

Z = 0.050
C = 2.500
B = 1.400
L = 0.020

$\lambda_1 = -4.1429 + 1.6843i$
$\lambda_2 = -4.1429 - 1.6843i$

L = 0.001
Z = 0.010
R = 1.000
C = 1.000

$\lambda_1 = -11.1268$
$\lambda_2 = -89.8732$

* Sistema presenta respuesta estable y subamortiguada en un escalón unitario de entrada

Subamortiguada

(1)

Error en estado estacionario:

* Se calcula en un escalón unitario de entrada

$$e(s) = \lim_{s \to 0} s P_a(s) \left(1 - \frac{P_p(s)}{P_a(s)}\right) = e(s) = \lim_{s \to 0} \frac{1}{s}\left(1 - \frac{LBs + Bz}{CLBz^2 + L(z+B)s + Bz}\right) = 0$$

(2)

Modelo de ecuaciones integro-diferenciales

$$F_z(t) F_L(t) = F_C(t) + F_B(t)$$

$$\frac{P_a(t)}{z} + \frac{1}{L}\int [P_a(t)]dt - C\frac{dP_p(t)}{dt} = \frac{P_p(t)}{B}$$

* Hipertenso

Z = 0.050
C = 2.500
B = 1.400
L = 0.020

$\lambda_1 = -4.1429 + 1.6843i$
$\lambda_2 = -4.1429 - 1.6843i$

Respuesta estable, subamortiguada en un escalón unitario de entrada

Error en estado

* Error en estado estacionario:

$$e(s) = \lim_{s \to 0} s P_a(s) \left(\frac{1 - P_F(s)}{P_a(s)} \right) = e(s) = \lim_{s \to 0} s \frac{1}{s} \left(1 - \frac{\frac{L}{R} + \frac{1}{2}}{CL\frac{1}{R}s^2 + \frac{L}{2}(2+2)s + \frac{1}{R}} \right)$$

$$\frac{1}{R} + \frac{1}{2}$$

$$1 + 0$$

* Modelo de ecuaciones íntegro-diferenciales:

Notemos se puede derivar funciones que se están derivando e integrando.

$$F_2(t) + F_L(t) = F_C(t) + F_R(t)$$

$$\frac{P_a(t)}{2} + \frac{1}{L}\int |P_a(t) - P_F(t)|\,dt - C\frac{dP_F(t)}{dt} = \frac{P_F(t)}{R} + \frac{P_F(t)}{2}$$

$$\left(\frac{1}{R} + \frac{1}{2}\right) P_F(t) = \frac{P_a(t)}{2} + \frac{1}{L}\int [P_a(t) - P_F(t)]\,dt - C\frac{dP_F(t)}{dt}$$

$$P_F(t) = \left[\frac{P_a(t)}{2} + \frac{1}{L}\int |P_a(t) - P_F(t)|\,dt - C\frac{dP_F(t)}{dt} \right] \frac{R_2}{2 - R_1}$$

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